CISC3014: Information Retrieval and Web Search

Tuesdays and Fridays; 4:00pm-5:15pm; venue: E11-1015, Fall 2026

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Office Hours: By appointment
TA: TBD
Course Platform: UMMoodle

Prerequisite(s)

- Fundamental algorithmic analysis and design of different programming constructs and data structures.

Students are expected to be comfortable with Python programming. Familiarity with linear algebra, probability, basic statistics, and machine-learning concepts such as training/validation/test splits will be helpful.

All textbooks, required readings, assignment specifications, and course announcements will be available through UMMoodle.

Text Books

- **[IR]** Christopher D. Manning, Prabhakar Raghavan, and Hinrich Schütze. *Introduction to Information Retrieval*. Cambridge University Press, 2008. Online edition.
- **[SE]** W. Bruce Croft, Donald Metzler, and Trevor Strohman. *Search Engines: Information Retrieval in Practice*. Pearson, 2010.
- **[RS]** Francesco Ricci, Lior Rokach, and Bracha Shapira, editors. *Recommender Systems Handbook*, 3rd edition. Springer, 2022.

Recommended Readings

- Jure Leskovec, Anand Rajaraman, and Jeffrey D. Ullman. *Mining of Massive Datasets*, 3rd edition. Cambridge University Press, 2020.
- Selected primary papers and current technical reports on sparse and dense retrieval, Learning to Rank, recommender systems, retrieval-augmented generation, and search agents. The required list will be posted on UMMoodle.

Course Description and Objectives

Information retrieval (IR) is the process that the computer system responds to users' queries on textual information, while web search is the application of IR on the largest web corpus (the web). IR system plays an essential role in our daily life, such as product recommendation, Advertising, social network, health care, intelligent education, smart city, etc. In this course, we will review the foundations of IR and web search, and introduce classical methods and models. After that, we will discuss recent advancement of Deep Learning in IR domain. This course is composed of course lectures, hands-on tutorials, projects, assignments and presentations. After this semester, you are expected to understand the key concepts of IR, and have the hands-on skills to solve real-world IR problems.

Programming Languages and Technical Environment

Python is the primary programming language. Students will use Jupyter notebooks and selected open-source libraries for retrieval, evaluation, vector search, recommendation, and language models. The teaching team will provide CPU-compatible baselines, fixed datasets, cached embeddings or model outputs where needed, and environment specifications. A paid API or personal GPU is not required for any core assessment.

Grading Policy

Homework assignments:	30%
Exam	50%
Project	20%
*Bonus	5%

The team project is the principal end-of-semester assessment. No separate written final examination is scheduled.

Grading Distribution

Condition 1:

Percentage Grade	Final Grade	Percentage Grade	Final Grade
– 93	A	92 - 88	A-
87 - 83	B+	82 - 78	B
77 - 73	B-	72 - 68	C+
67 - 63	C	62 - 58	C-
57 - 53	D+	52 - 50	D
Below 50	F		

Condition 2:

Grades	Maximum Cumulative Percentage of Passing Students
A	10%
A- or above	25%
B- or above	75%
C or above	100%
D or above	100%

***The final grade distribution will satisfy both condition 1 and condition 2. Final grades in the course may be curved at the instructor's discretion.**

Homework Policy

- Homework will be due one or two weeks after assignment unless otherwise noted, no late homework is accepted unless you can provide official notice (e.g., academic conference attendance notice, medical emergency notice, etc) to the instructor in advance.

- Possible revision of homework grades may be discussed with the grader within one week from the return of the marked homework.
- Homework will be composed of calculation, concept understanding and programming assignments.
- This course does not accept last-minute emails about why submissions are late (e.g., internet issues, website issues, laptop issues, UMMoodle issues).
- According to the university honor code, it is your responsibility to start earlier, to complete earlier, to submit on time, and to finish independently. Plagiarism strictly prohibited. A score of zero will be assigned to the homework to all involved parties once plagiarism is found. Two times of plagiarism will result in failure of the course, and will be reported to the university for further action.

Project Policy

- Students will work in group (no more than two students) for the course project.
- The topics and requirements will be announced and discussed in class. Self-proposal is welcomed.
- Project should be composed of demos, presentations and reports.
- The project will be presented at the end of semester.

Exam Policy

- One mid-term exam will be held during the semester.
- One final exam will be held at the end of the semester.
- Students should comply with the honor code to avoid plagiarism. Students should finish the exam independently. Plagiarism in exam will result in failure of the course, and will be reported to the university for further action.
- In order to make up a missed exam, the student should provide the instructor with sufficient evidence to justify the absence.

LLM Use and Academic Integrity

The use of an LLM is permitted in homework and project work only when the assessment instructions allow it. Students must disclose the model or tool, version when available, purpose, and important prompts or generation steps. Students remain responsible for every submitted line of code, numerical result, citation, and written claim. Unverified model output must not be presented as a source or an experimental result.

Students must not upload personal data, restricted datasets, unpublished course solutions, or other unauthorized material to an external service. Course agents must run with limited tools and explicit step limits; they may not autonomously send messages, make purchases, delete data, or modify real accounts. The mid-term examination does not permit LLM use.

Plagiarism, unauthorized collaboration, fabricated results, and misrepresentation of AI-generated work will be handled under the university's academic-integrity rules. Project presentations may include an individual defense, code inspection, and checks against fixed outputs to confirm that each student understands the submitted work.

Schedule (Non-Restrictive)

Topics and readings may be adjusted to reflect class progress, university closures, or changes in the technical environment. Changes to assessment weights or deadlines will follow university rules and will be announced through UMMoodle.

Week 1 (18/08, 21/08)

- Introduction: search, recommendation, RAG, and agents as information-access systems.
- Classical IR Part 1: Boolean Retrieval Model (Index Construction and Compression)

Week 2 (25/08, 28/08)

- Classical IR Part II: Vector Space Model (Scoring, Term Weighting)
- IR evaluation and experimental design: qrels, MAP, MRR, nDCG, and Recall@k

Week 3 (01/09, 04/09)

- Link analysis
- Web crawling I

Week 4 (08/09, 11/09)

- Web crawling II

Week 5 (15/09, 18/09)

- Neural retrieval foundations I: machine learning basics.
- Neural retrieval foundations II: Transformers, bi-encoders, cross-encoders, and contrastive.

Week 6 (22/09, 25/09)

- Approximate nearest-neighbor search, hybrid retrieval, reciprocal-rank fusion, and late interaction.
- Learning to Rank and user feedback: pointwise, pairwise, and listwise approaches.

Week 7 (29/09, 02/10)

- Recommender systems I: collaborative filtering, implicit feedback, and matrix factorization.

Week 8 (06/10, 09/10)

- Recommender systems II: multi-stage recommendation, sequential behavior, cold start, exposure bias, diversity, and fairness.

Week 9 (13/10, 16/10)

- Mid-Term Examination.
- Learned sparse and dense retrieval; hard-negative mining.

Week 10 (20/10, 23/10)

- **20/10** LLMs for IR and generative retrieval: query, candidate, ranking, and presentation roles.
- **23/10** RAG I: corpus design, chunking, indexing, retrieval, and context packing.

Week 11 (27/10, 30/10)

- **27/10** RAG II: hybrid retrieval, query rewriting, multi-hop questions, and iterative retrieval.
- **30/10** RAG III: component-wise evaluation, faithfulness, citation, and abstention.

Week 12 (03/11, 06/11)

- **03/11** Search agents I: task decomposition, tools, state, traces, and stopping conditions. Project milestone check.

- **06/11** Search agents II: deep research, evaluation, safety, prompt injection, and human control. Assignment 3 due; artifact-freeze check.

Week 13 (10/11, 13/11)

- **10/11** Project Presentation A.
- **13/11** Project Presentation B.

Week 14 (17/11, 20/11)

- **17/11** Project Presentation C.
- **20/11** Project Presentation D.

Week 15 (24/11, 27/11)

- **24/11** Project Presentation E.
- **27/11** Project fair, remaining presentations, and course synthesis.